

GSS SERVER SHLS 700

**Server-based,
real-time encoding
and transcoding
system**



Hardware base SHLS 710/720

PRODUCT DESCRIPTION

The SHLS 7XX series is the compact and efficient solution for network operators, hotels and hospitals that want to protect their existing infrastructure and provide modern streaming formats at the same time.

With high-density transcoding, many streams can be processed in parallel – ideal for IPTV and hospitality environments.

Thanks to investment protection for existing systems, HLS/OTT compatibility and multi-device support, SHLS enables reliable and future-proof signal processing.

This allows network operators to remain flexible while meeting the requirements of entertainment, mobile devices and smart TVs.

FEATURES

Transcoder

- Real-time transcoding
- Input SPTS/MPTS (UDP/RTP; CBR/VBR)
- Output SPTS (UDP/RTP; CBR)
- Operating mode "UHD → HD"*:
Up to 10 services can be simultaneously transcoded e.g. from HEVC (H.265) to MPEG-4 (H.264)
- Operating mode "HD → SD"*:
Up to 40 services can be simultaneously transcoded e.g. from MPEG-4 (H.264) to MPEG-2 (H.262)
- Perfect for network operators and TV providers: full pass-through of all services - EPG, teletext, audio tracks and all associated metadata
- Optional: Adaptive LL-HLS transcoding (SHLS 700 Pro)

*(depending on the hardware SHLS710/720)

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FEATURES

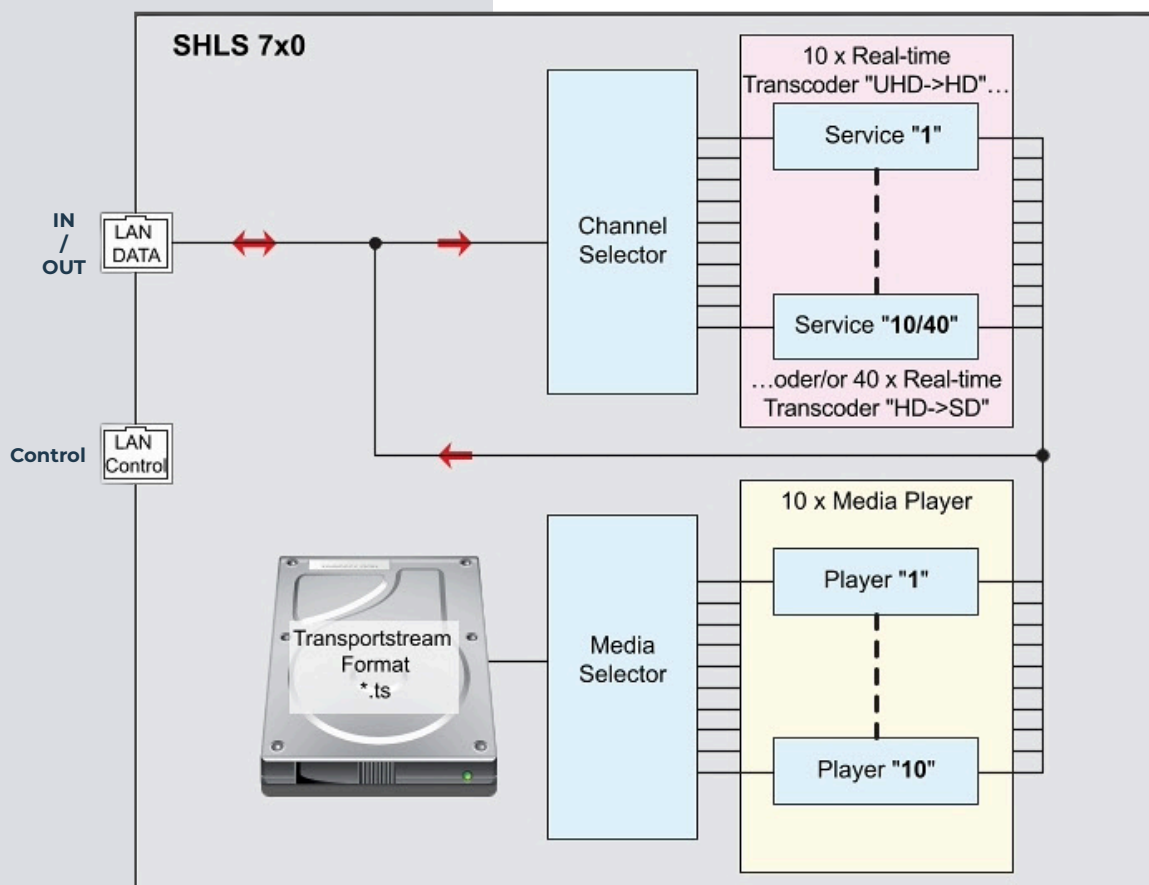
Media player

- 10 streams simultaneously
- Playback of files in *.ts format
- File upload via the operating software or SFTP access

Converter for Media player

- Supported input formats
mkv, avi, divx, mp4, mov and mpeg are supported.
- Output format ts.

HTML user interface



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TECHNICAL DATA

The product fulfils the guidelines and standards for CE labelling.
 Unless otherwise noted all values are specified as "typical".

Stream output SPTS	
Number of streams Media Player	10
Number of streams Transcoder (SHLS 710)	10 UHD → HD or 20 HD → SD
Number of streams Transcoder (SHLS 720)	10 UHD → HD or 40 HD → SD
Protocols	UDP/RTP
Input SPTS/MPTS (UDP/RTP)	
Video formats	MPEG-2 (H.262); MPEG-4 (H.264); HEVC (H.265)
Resolutions	720x576; 1280x720; 1920x1080;
Frame rate	25i; 25p, 30p; 50p; 60p;
Audio formats	MP2; MP3; AAC
Output SPTS (UDP)	
Video formats	MPEG-2 (H.262); MPEG-4 (H.264); HEVC (H.265)
Resolutions	720x576; 1280x720; 1920x1080;
Frame rate	25i; 25p, 30p; 50p; 60p;
Bit rate	CBR
Audio formats	MPEG-1 Audio Layer 2 (MP2); MPEG-1 Audio Layer 3 (MP3); Audio Coding 3 (AC-3)
Media Player	
File format for Media Player	Transport Stream (*.ts)
Server Hardware	
1U chassis	supermicro CSE-815TQ-R500WB
Power supply (redundant)	100-240V 50/60 Hz; 500W 80PLUS Platinum (94%) Certifications: BSMI, CCC, CE/EMC; FCC class A; TUV/CB; UL/CUL
Dimensions (19" / 1HE)	43 mm (H), 437 mm (W), 650 mm (D)
Weight	16.33 kg
Ambient temperature	5 - 35 °C
Air humidity (non-condensing)	8% - 90 %
LAN interface (for Data)	
Standard	1000-BASE-T
Data rate	≤ 800 MBit/s
Connections	
LAN 1000-BASE-T	2 x RJ45
USB sockets	2 x USB A 3.0; 2 x USB A 2.0

GSS SERVER SHLS 700 Pro

PRESETS AND BENCHMARKING

The product fulfils the guidelines and standards for CE labelling.

Unless otherwise noted all values are specified as "typical".

Video codec, resolution, data rates default settings (not changeable):

HLS VIDEO PRESETS			
Video			
Codec	Description	Resolution	Data Rate
H.264	240p	426x240	400 kbit/s
H.264	240p	426x240	600 kbit/s
H.264	360p	640x360	700 kbit/s
H.264	360p	640x360	900 kbit/s
H.264	432p	768x432	800 kbit/s
H.264	432p	768x432	900 kbit/s
H.264	480p	854x480	900 kbit/s
H.264	480p	854x480	1100 kbit/s
H.264	576p	1024x576	1250 kbit/s
H.264	576p	1024x576	1600 kbit/s
H.264	720p	1280x720	2500 kbit/s
H.264	720p	1280x720	3200 kbit/s
H.264	Full HD 1080p	1920x1080	4500 kbit/s
H.264	Full HD 1080p	1920x1080	5300 kbit/s
Frame Rate			
Supported FPS		25, 30, 50, 60	
Audio			
Audio-Codec		Data Rates	
MP3		32–320 kbit/s	
AAC		32–320 kbit/s	
AC-3		32–320 kbit/s	
BENCHMARKING OUTPUTS PER INPUT (WITH 2 GPUS SHLS 720 HARDWARE)			
Input: DVB-T2 (HEVC, 1920×1080) → 3 output streams			
Output Stream		Resolution	Data Rate
Stream 1		1280x720	3.2 Mbit/s
Stream 2		768x576	900 kbit/s
Stream 3		640x360	900 kbit/s
Approximate number of programs: approx. 10			
Input: DVB-S2 (MPEG-4, 1280×720) → 3 output streams			
Output Stream		Resolution	Data Rate
Stream 1		1280x720	3.2 Mbit/s
Stream 2		768x576	900 kbit/s
Stream 3		640x360	900 kbit/s
Approximate number of programs: approx. 12			
Input: DVB-S2 (MPEG-2, 720×576) → 3 output streams			
Output Stream		Resolution	Data Rate
Stream 1		1280x720	3.2 Mbit/s
Stream 2		768x576	900 kbit/s
Stream 3		640x360	900 kbit/s
Approximate number of programs: approx. 10			